

International Year One Engineering

Computer Programming in C

Exercises 3. Conditional Statements

Text in this font is the description of the task.

Text in this font is sample output of a program to the screen.
Example keyboard input to the screen appears as underlined e.g. 34

1. Write a program to request a number and then print whether the number is a positive number or a negative number.

Enter number: -567

The number is negative

2. Write a program to request a user to input a single character y or Y to indicate a yes response and an n or N to indicate a no response and print the response as a word YES or NO.

Do you want to continue (y/n) or (Y/N)?: N

You responded NO

3. Write a conditional expression using the ? operator to assign the value of 1 to a variable if it is greater than 50 and to assign it to 0 if its value is less than 50.

Enter value: 38

Variable reset to 0

4. Write a program that reads a grade in the range 'A' to 'E' as either upper case or lower case. Then implement a switch statement that prints out the following text remarks to describe the grades.

Grade	Remark
A	Excellent
B	Good
C	Average
D	Poor
E	Failing

Enter Grade: B

Your performance was Good

Projects 3. Conditional Statements

1. Write a program that converts a temperature value between degrees Fahrenheit and degrees Celsius and vice versa. The program should allow you to:

- input the value of temperature to be converted.
- indicate which conversion is to be performed, by inputting F or f to convert from Celsius to Fahrenheit, or inputting C or c to convert from Fahrenheit to Celsius

The conversion formulae from Fahrenheit to Celsius and vice versa are given by:

- Fahrenheit = Celsius $\times$ 9/5 + 32
- Celsius = (Fahrenheit - 32) $\times$ 5/9

Enter temperature: 30

Enter conversion required (f or F to convert to Fahrenheit, c or C to convert to Celsius): F

30 degrees Celsius is equal to 86 degrees Fahrenheit

2. Write a program in which a user is requested to enter the first three letters of a day and then outputs the day number according to following table. The program should be case-insensitive i.e. the user may input any combination of uppercase and lowercase letters and the program will still work.

Day	Number
Mon	1
Tue	2
Wed	3
Thu	4
Fri	5
Sat	6
Sun	7

Enter a day (Mon/Tue/Wed/Thu/Fri/Sat/Sun) : Wed

The day number is : 3

3. Write a program which allows a user to input two integer numbers and determines which of the following conditions applies:

- (A) Both numbers are even (divisible by 2)
- (B) Both numbers are odd (not divisible by 2)
- (C) One number is even and the other number is odd

Enter two integer numbers : 24 37

Condition C is true

4. A component manufacturer offers the following discounts on different order amounts for different product codes, as shown in the table below. Write a program that requests the product code and the number of units ordered and hence calculates the total order price based on the discount table.

		Quantity			
Code	Unit Price	1-10	10-49	50-99	100+
W	12.45	0%	2%	5%	7.5%
X	2.36	0%	0.5%	2%	4%
Y	6.50	0%	1%	2.5%	5%
Z	16.40	0%	10%	12.5%	15%

Your program should produce input and output like the following:

Product Code : Y
Order Amount : 120

Total Cost before discount : 780
Total Cost after discount : 741